PTvsFT

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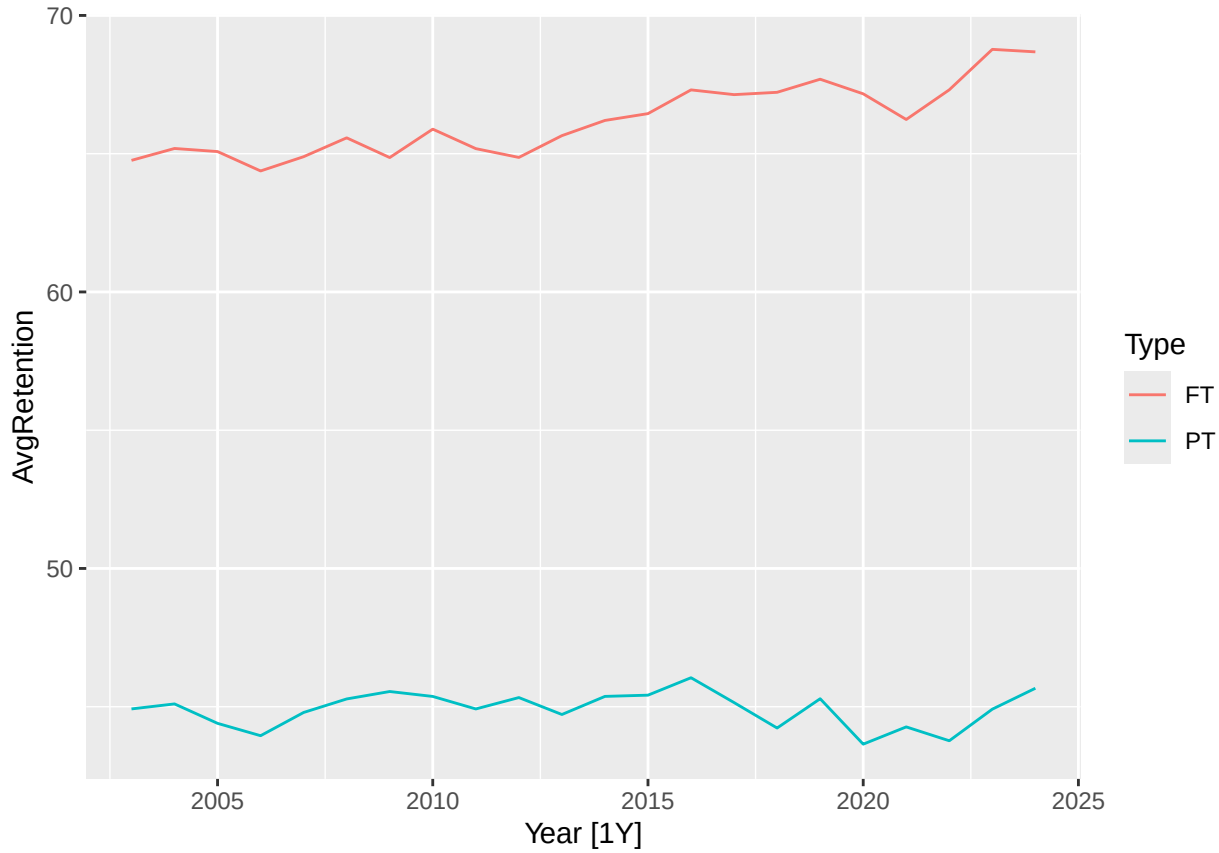
2026-04-28

```r
# Load required packages
library(tidyverse)
library(lubridate)
library(tsibble)
library(feasts)
library(fable)
```

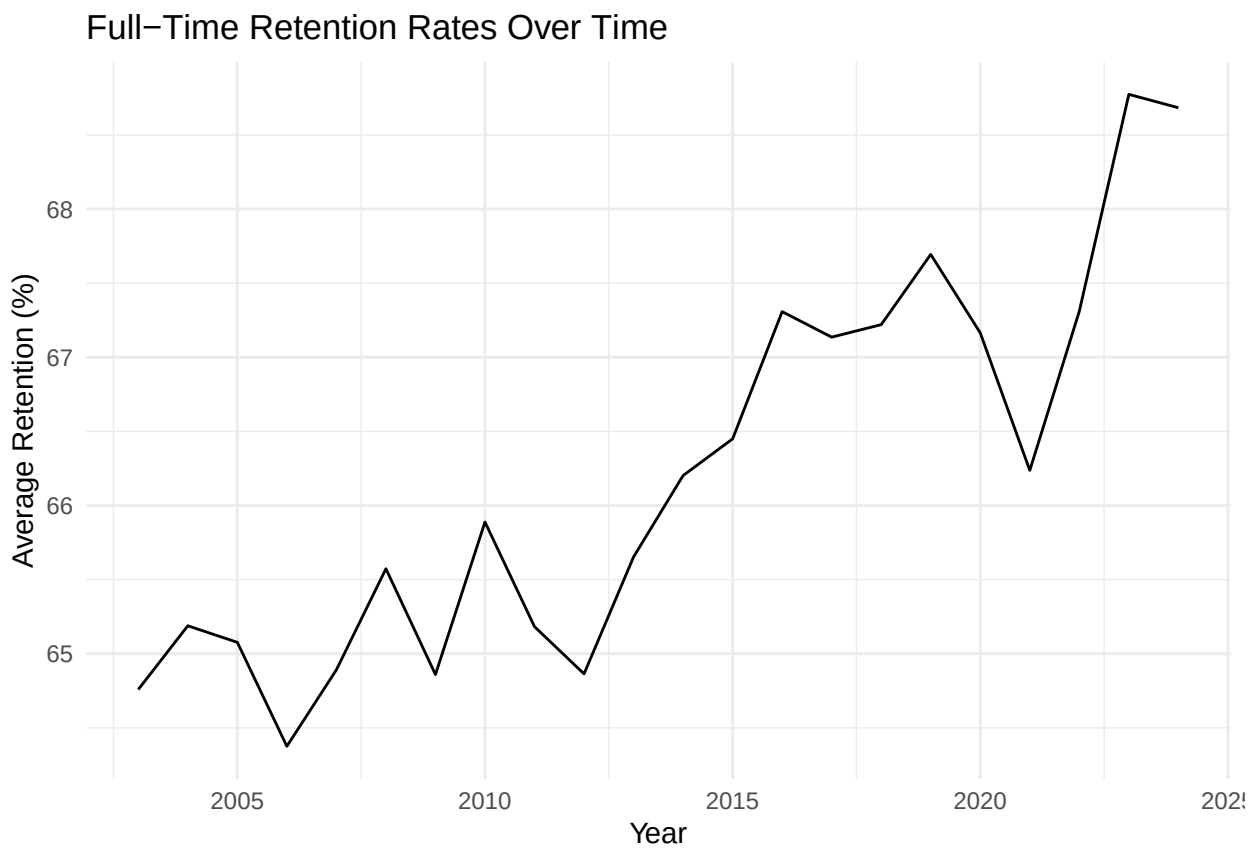
```r
# Choose file interactively
file_path <- file.choose()

# Load data
df <- read_csv(file_path)

# Reshape wide → long
df_long <- df %>%
  pivot_longer(
    cols = -ID,
    names_to = c("Year", "Type"),
    names_sep = "_",
    values_to = "Retention"
  ) %>%
  mutate(
    Year = as.integer(Year),
    Type = factor(Type)
  )

# Aggregate BEFORE converting to tsibble
df_yearly <- df_long %>%
  group_by(Year, Type) %>%
  summarise(AvgRetention = mean(Retention, na.rm = TRUE)) %>%
  ungroup()

# Now convert to tsibble (no errors)
df_yearly_ts <- df_yearly %>%
  as_tsibble(index = Year, key = Type)

# Quick plot
df_yearly_ts %>%
  autoplot(AvgRetention)
```



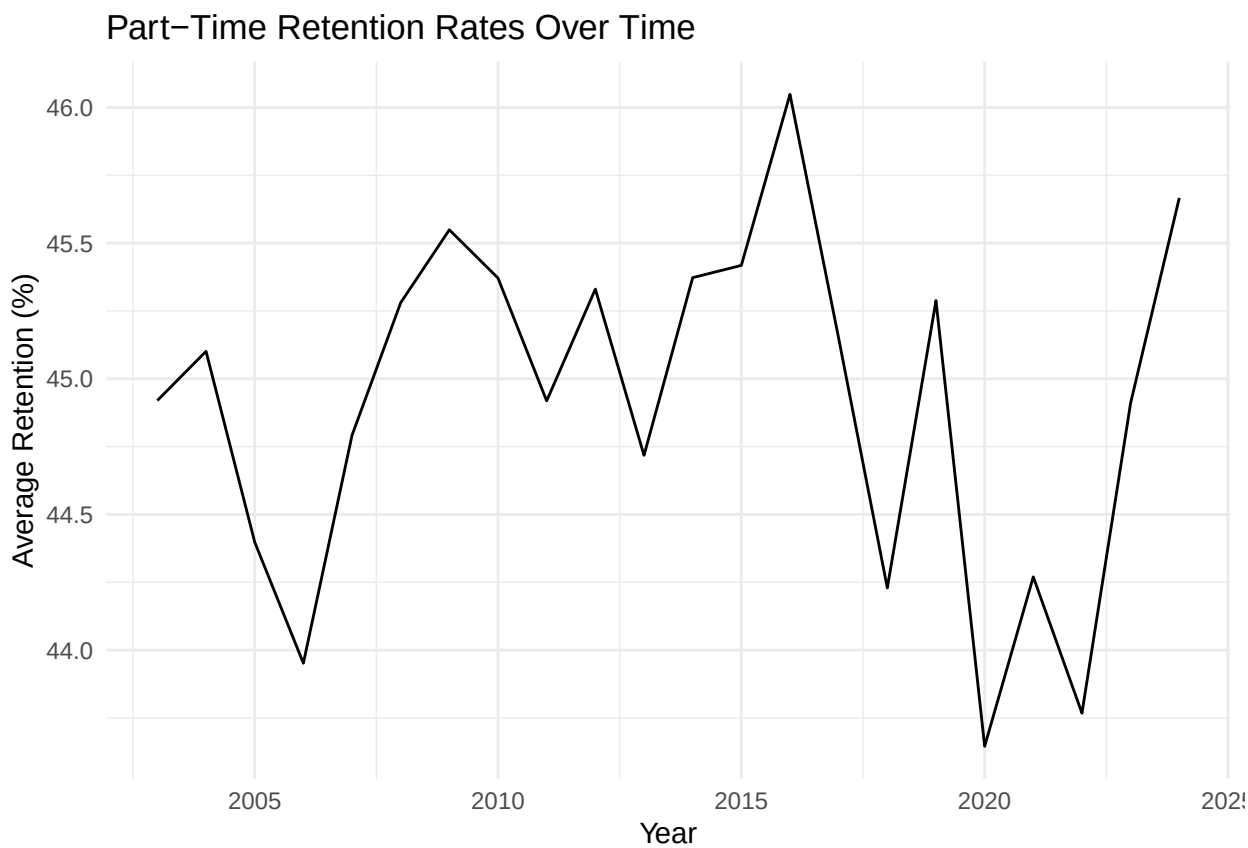

```r
# Example ARIMA model
fit <- df_yearly_ts %>%
  model(ARIMA(AvgRetention))

report(fit)
```

```
## # A tibble: 2 x 9
##   Type  .model              sigma2 log_lik   AIC  AICc   BIC ar_roots  ma_roots
##   <fct> <chr>               <dbl>   <dbl> <dbl> <dbl> <dbl> <list>    <list>
## 1 FT    ARIMA(AvgRetention)  0.477   -22.0  46.0  46.3  47.1 <cpl [0]> <cpl [0]>
## 2 PT    ARIMA(AvgRetention)  0.406   -20.8  45.6  46.2  47.7 <cpl [0]> <cpl [0]>
```

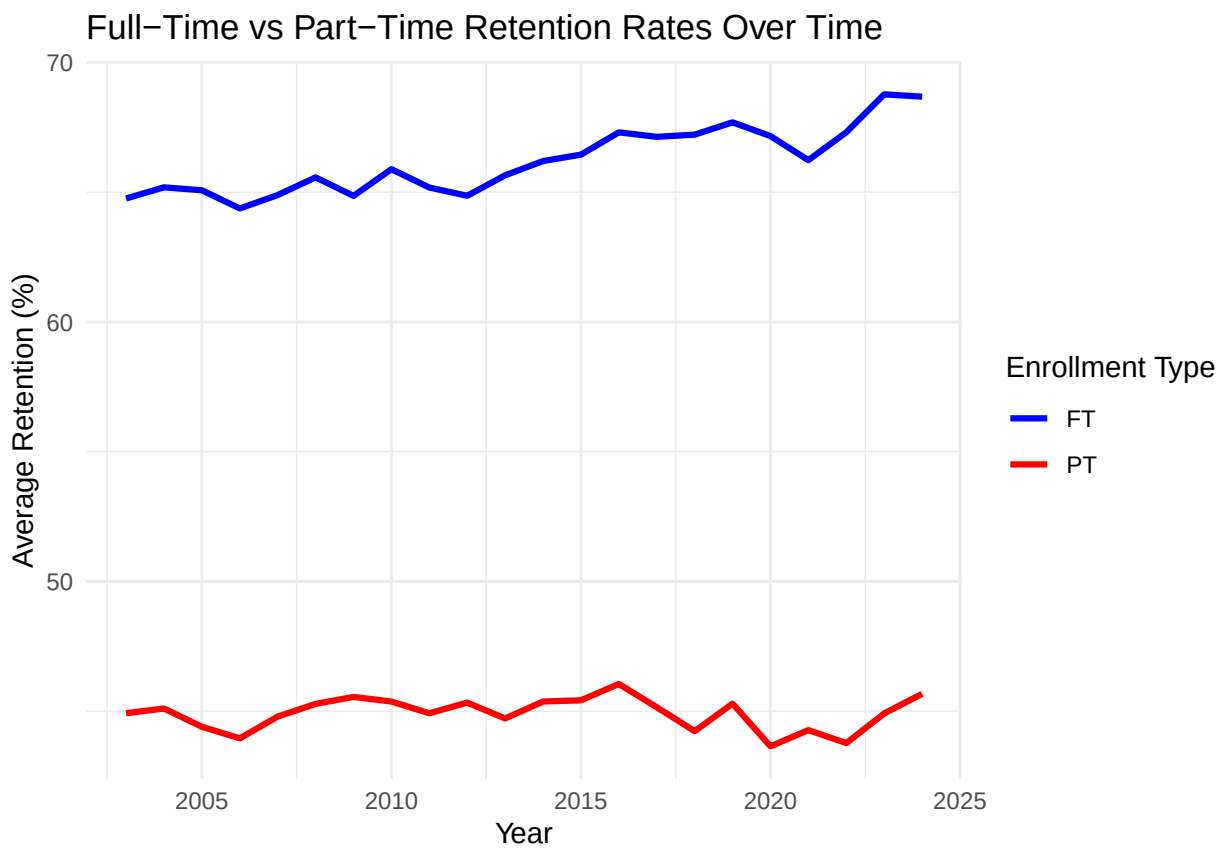
```r
# FULL-TIME RETENTION TIME SERIES
df_yearly_ts %>%
  filter(Type == "FT") %>%
  autoplot(AvgRetention) +
  labs(
    title = "Full-Time Retention Rates Over Time",
    x = "Year",
    y = "Average Retention (%)"
  ) +
  theme_minimal()
```



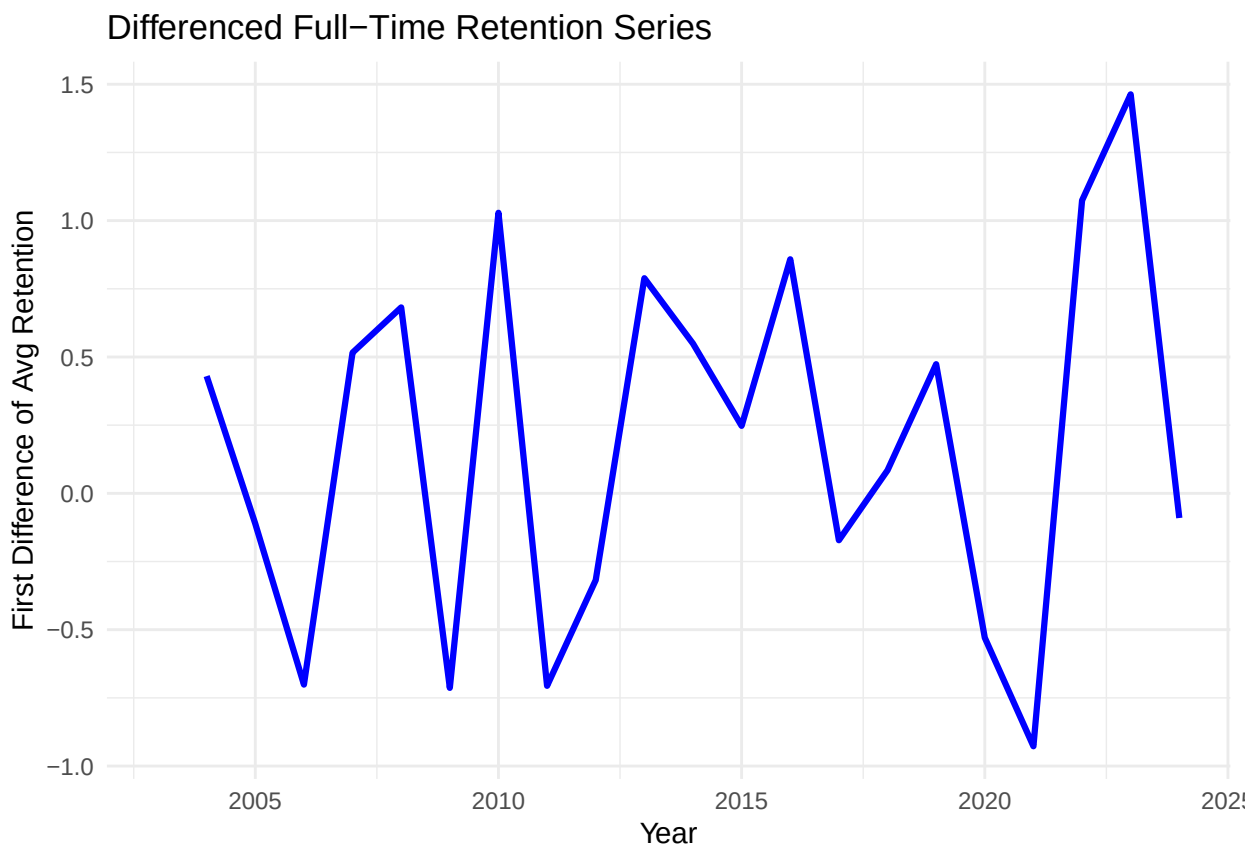

```r
# PART-TIME RETENTION TIME SERIES
df_yearly_ts %>%
  filter(Type == "PT") %>%
  autoplot(AvgRetention) +
  labs(
    title = "Part-Time Retention Rates Over Time",
    x = "Year",
    y = "Average Retention (%)"
  ) +
  theme_minimal()
```



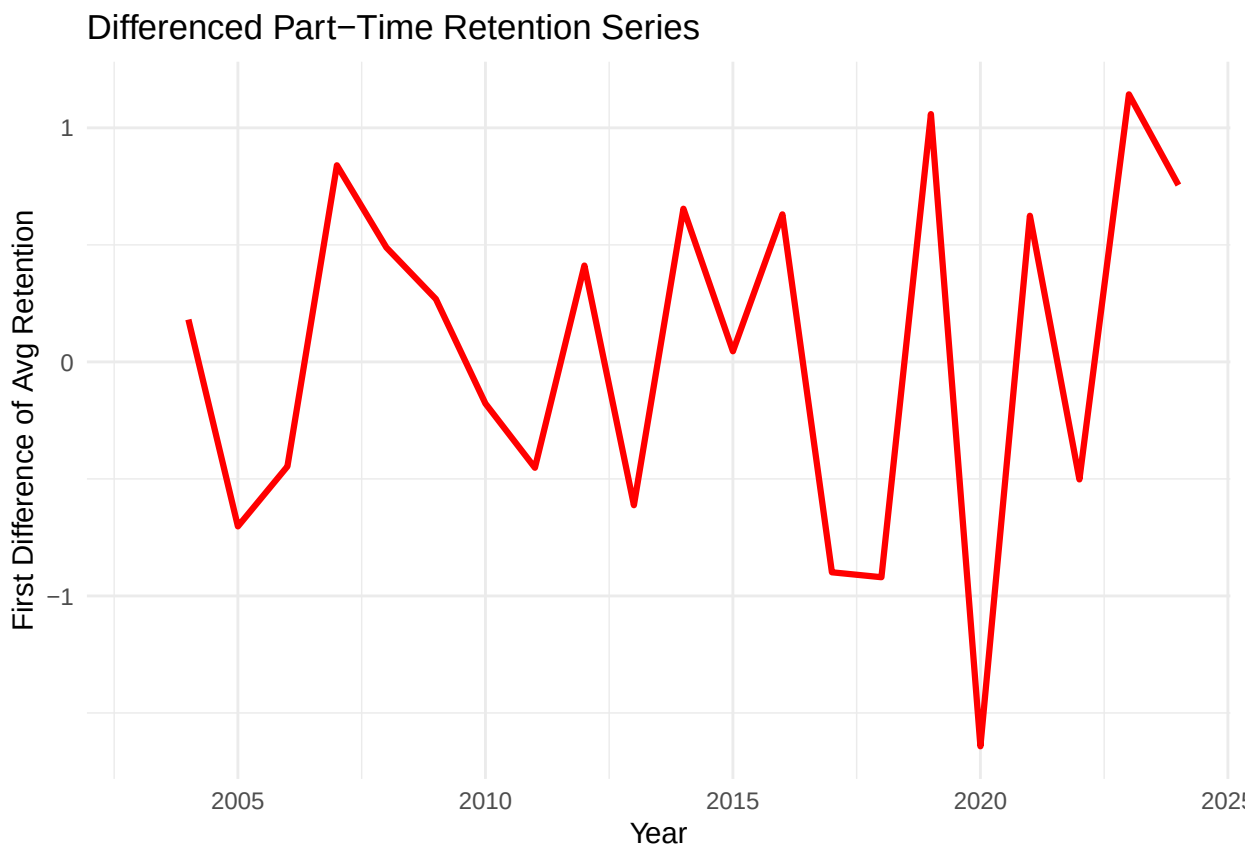

```
df_yearly_ts %>%
  ggplot(aes(x = Year, y = AvgRetention, color = Type)) +
  geom_line(size = 1.1) +
  labs(
    title = "Full-Time vs Part-Time Retention Rates Over Time",
    x = "Year",
    y = "Average Retention (%)",
    color = "Enrollment Type"
  ) +
  scale_color_manual(values = c("FT" = "blue", "PT" = "red")) +
  theme_minimal()
```



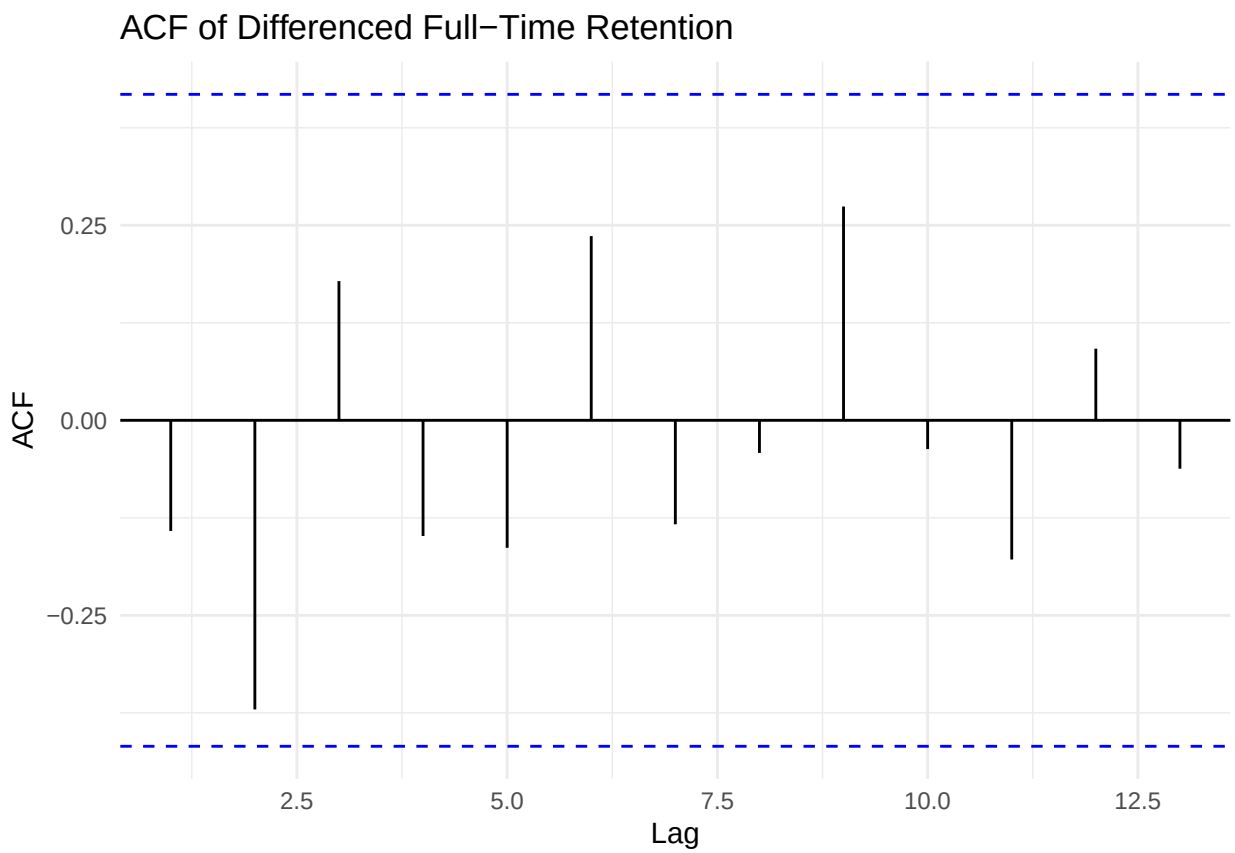

```r
# DIFFERENCED FULL-TIME SERIES
df_yearly_ts %>%
  filter(Type == "FT") %>%
  mutate(Diff = difference(AvgRetention)) %>%
  ggplot(aes(x = Year, y = Diff)) +
  geom_line(color = "blue", size = 1.1) +
  labs(
    title = "Differenced Full-Time Retention Series",
    x = "Year",
    y = "First Difference of Avg Retention"
  ) +
  theme_minimal()
```



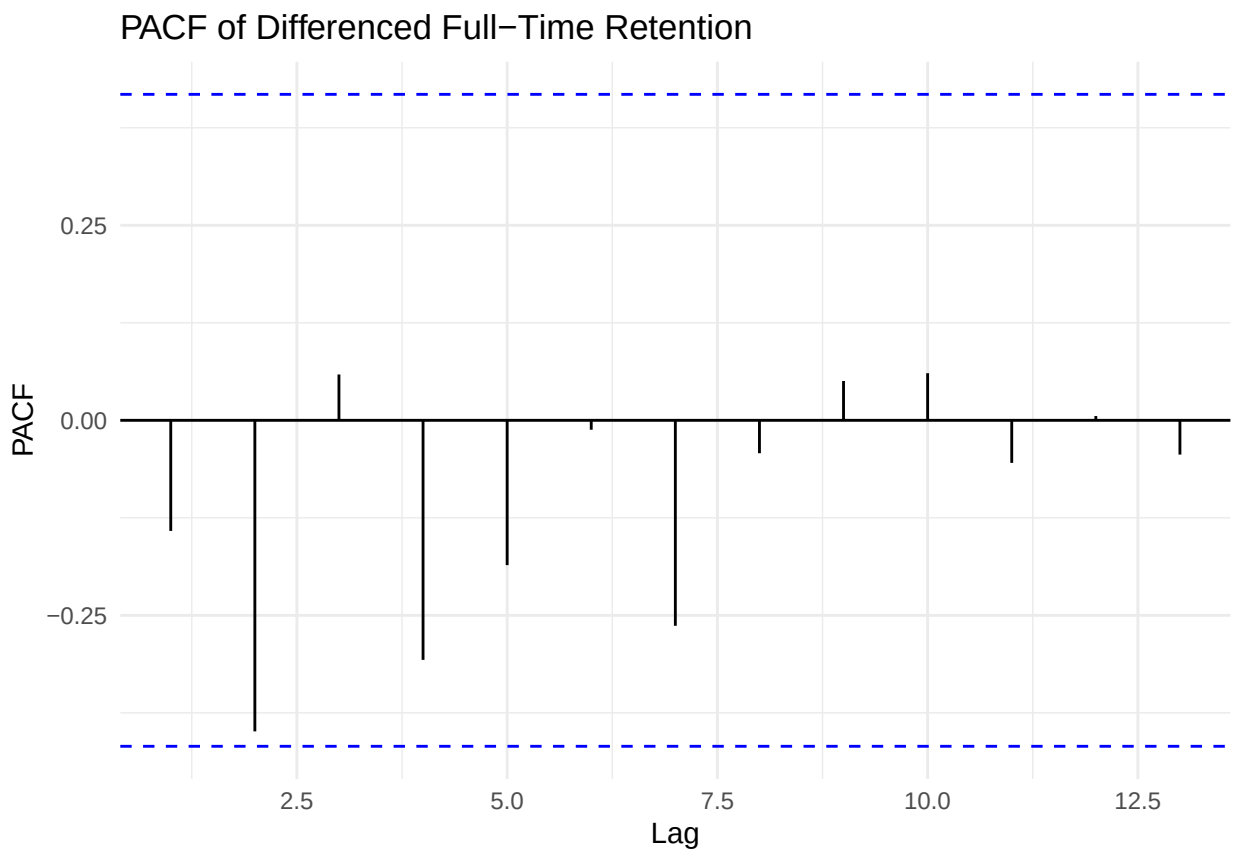

```r
# DIFFERENCED PART-TIME SERIES
df_yearly_ts %>%
  filter(Type == "PT") %>%
  mutate(Diff = difference(AvgRetention)) %>%
  ggplot(aes(x = Year, y = Diff)) +
  geom_line(color = "red", size = 1.1) +
  labs(
    title = "Differenced Part-Time Retention Series",
    x = "Year",
    y = "First Difference of Avg Retention"
  ) +
  theme_minimal()
```



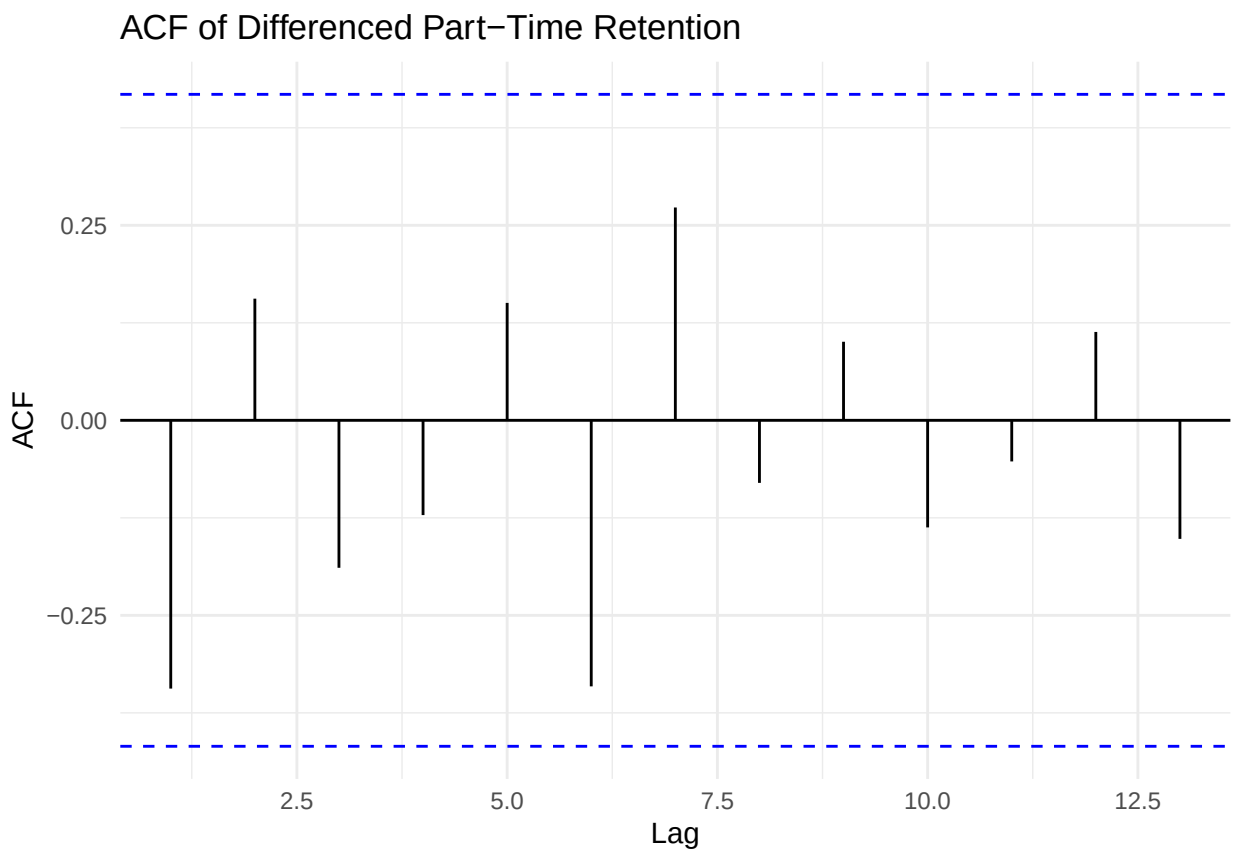

```r
# DIFFERENCED FULL-TIME SERIES
ft_diff <- df_yearly_ts %>%
  filter(Type == "FT") %>%
  mutate(Diff = difference(AvgRetention))

# ACF
ft_diff %>%
  ACF(Diff) %>%
  autoplot() +
  labs(
    title = "ACF of Differenced Full-Time Retention",
    x = "Lag",
    y = "ACF"
  ) +
  theme_minimal()
```



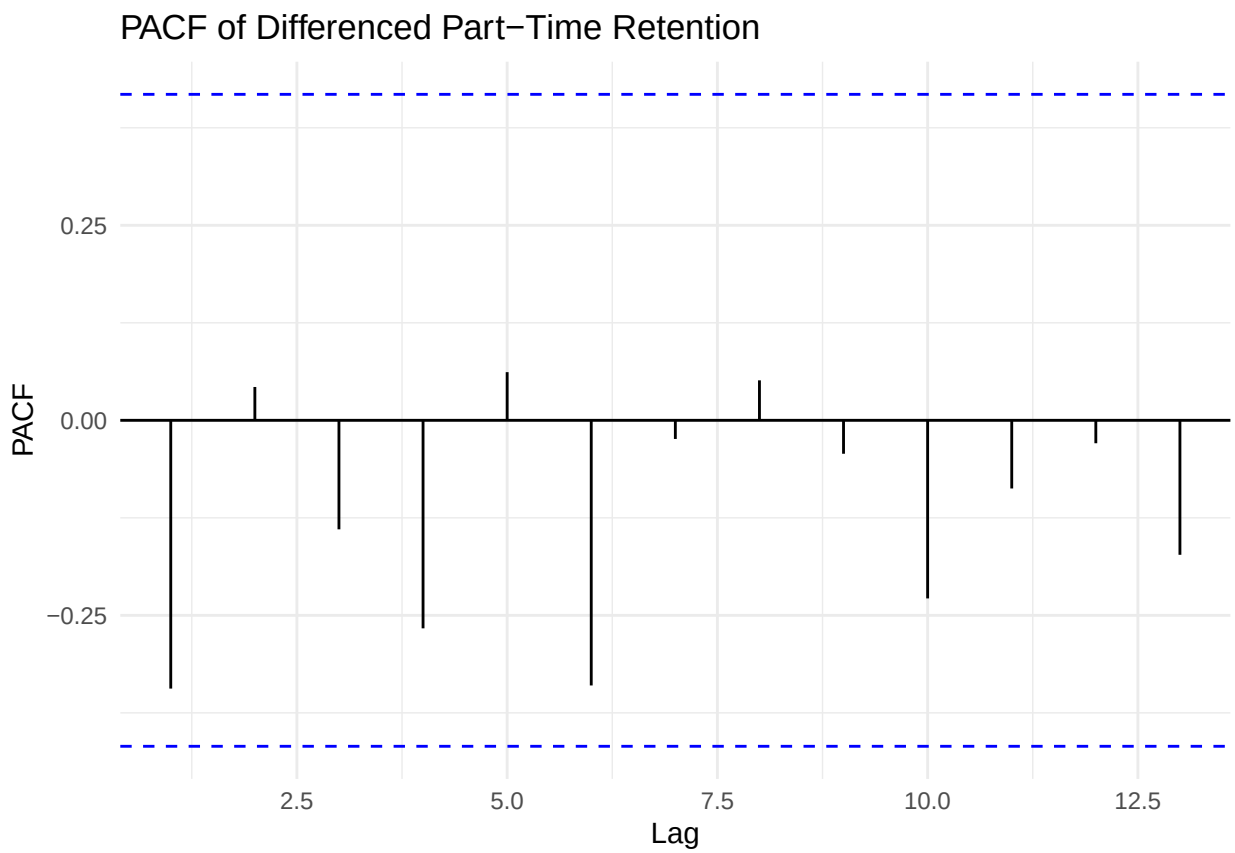

```r
# PACF
ft_diff %>%
  PACF(Diff) %>%
  autoplot() +
  labs(
    title = "PACF of Differenced Full-Time Retention",
    x = "Lag",
    y = "PACF"
  ) +
  theme_minimal()
```




```r
# DIFFERENCED PART-TIME SERIES
pt_diff <- df_yearly_ts %>%
  filter(Type == "PT") %>%
  mutate(Diff = difference(AvgRetention))

# ACF
pt_diff %>%
  ACF(Diff) %>%
  autoplot() +
  labs(
    title = "ACF of Differenced Part-Time Retention",
    x = "Lag",
    y = "ACF"
  ) +
  theme_minimal()
```




```r
# PACF
pt_diff %>%
  PACF(Diff) %>%
  autoplot() +
  labs(
    title = "PACF of Differenced Part-Time Retention",
    x = "Lag",
    y = "PACF"
  ) +
  theme_minimal()
```




```r
# FT model
fit_ft <- df_yearly_ts %>%
  filter(Type == "FT") %>%
  model(ARIMA(AvgRetention))

report(fit_ft)
```

```
## Series: AvgRetention
## Model: ARIMA(0,1,0)
##
## sigma^2 estimated as 0.477:  log likelihood=-22.02
## AIC=46.04   AICc=46.25   BIC=47.08
```

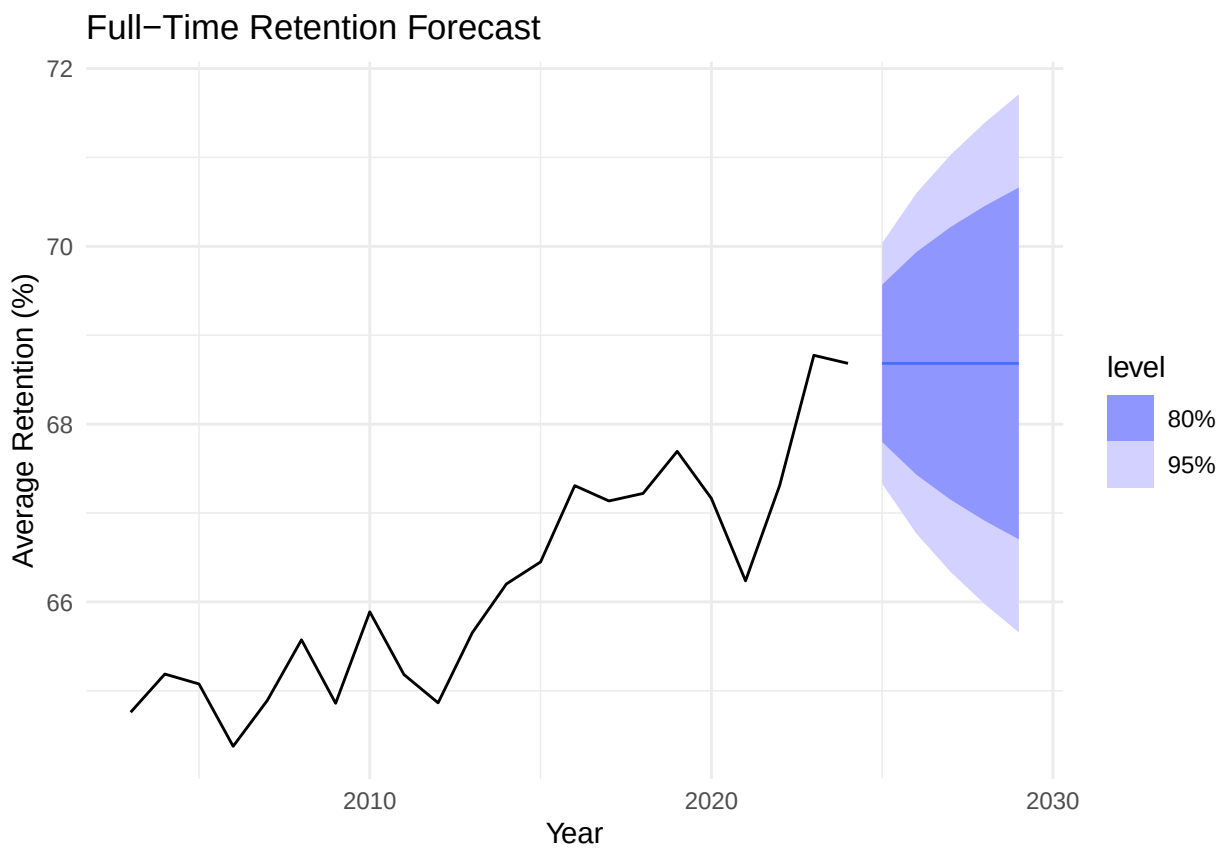
```r
# PT model
fit_pt <- df_yearly_ts %>%
  filter(Type == "PT") %>%
  model(ARIMA(AvgRetention))

report(fit_pt)
```

```
## Series: AvgRetention
## Model: ARIMA(0,0,0) w/ mean
##
## Coefficients:
##       constant
```

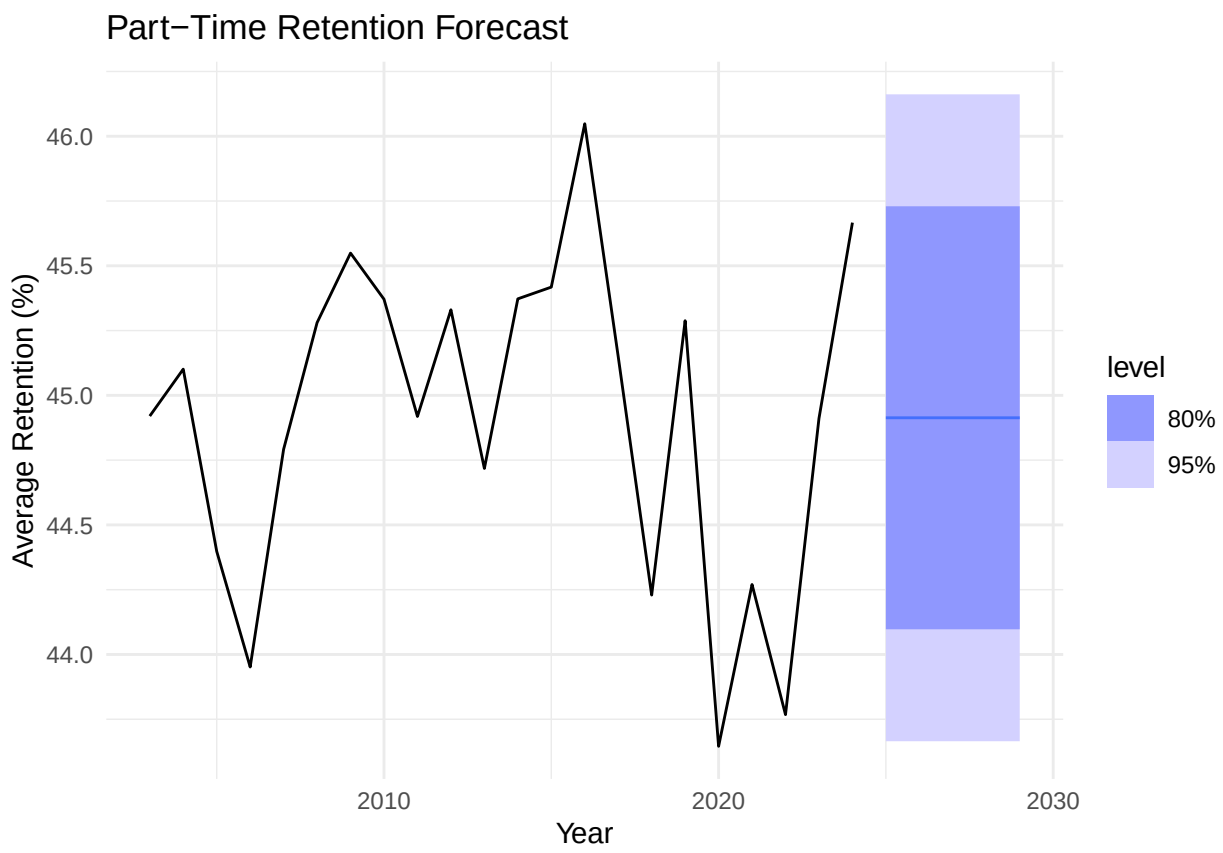
```
##        44.9135
## s.e.    0.1327
##
## sigma^2 estimated as 0.4057:  log likelihood=-20.78
## AIC=45.56   AICc=46.19   BIC=47.74
```

```r
# FT forecast
fc_ft <- fit_ft %>%
  forecast(h = 5)

# PT forecast
fc_pt <- fit_pt %>%
  forecast(h = 5)
```

```r
# FT model
fit_ft <- df_yearly_ts %>%
  filter(Type == "FT") %>%
  model(ARIMA(AvgRetention))

# FT forecast
fc_ft <- fit_ft %>%
  forecast(h = 5)

# FT forecast plot
fc_ft %>%
  autoplot(df_yearly_ts %>% filter(Type == "FT")) +
  labs(
    title = "Full-Time Retention Forecast",
    x = "Year",
    y = "Average Retention (%)"
  ) +
  theme_minimal()
```

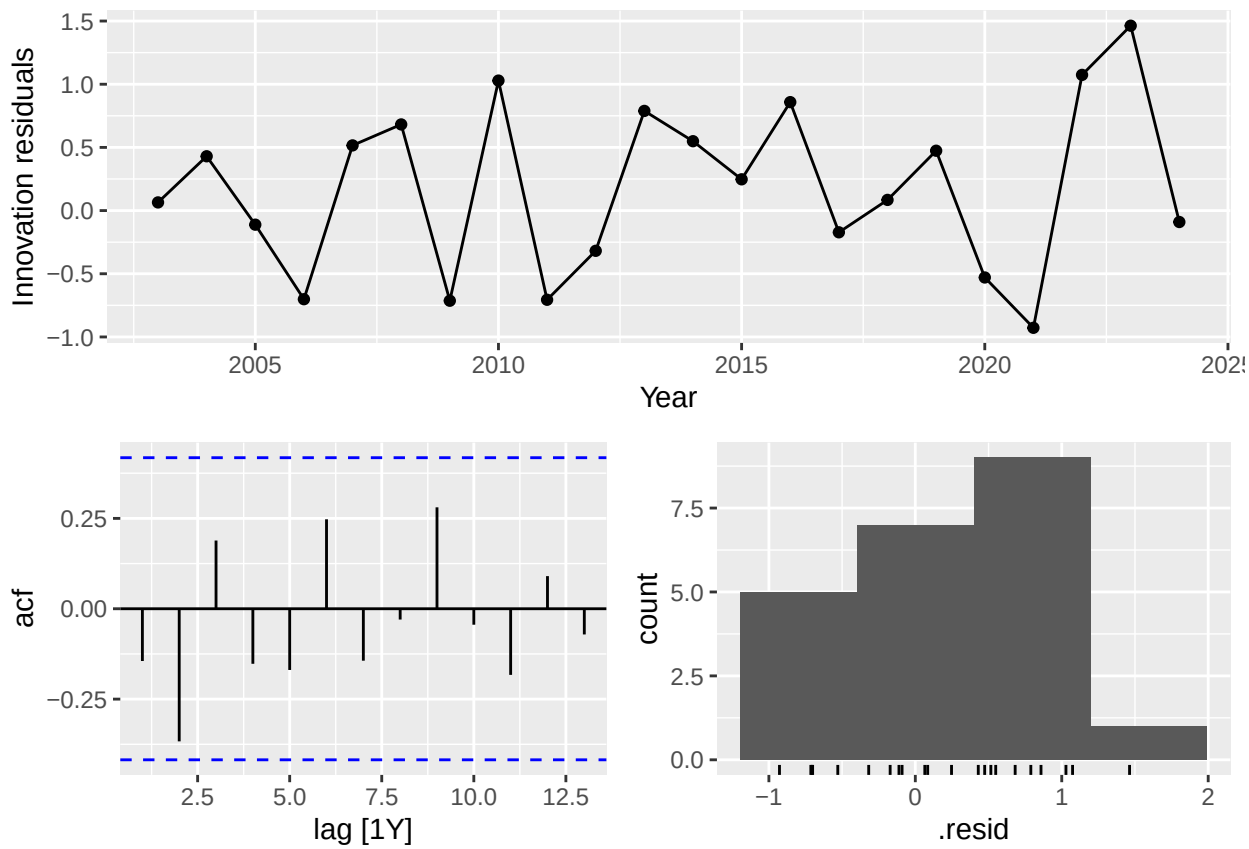



```r
# PT model
fit_pt <- df_yearly_ts %>%
  filter(Type == "PT") %>%
  model(ARIMA(AvgRetention))

# PT forecast
fc_pt <- fit_pt %>%
  forecast(h = 5)

# PT forecast plot
fc_pt %>%
  autoplot(df_yearly_ts %>% filter(Type == "PT")) +
  labs(
    title = "Part-Time Retention Forecast",
    x = "Year",
    y = "Average Retention (%)"
  ) +
  theme_minimal()
```

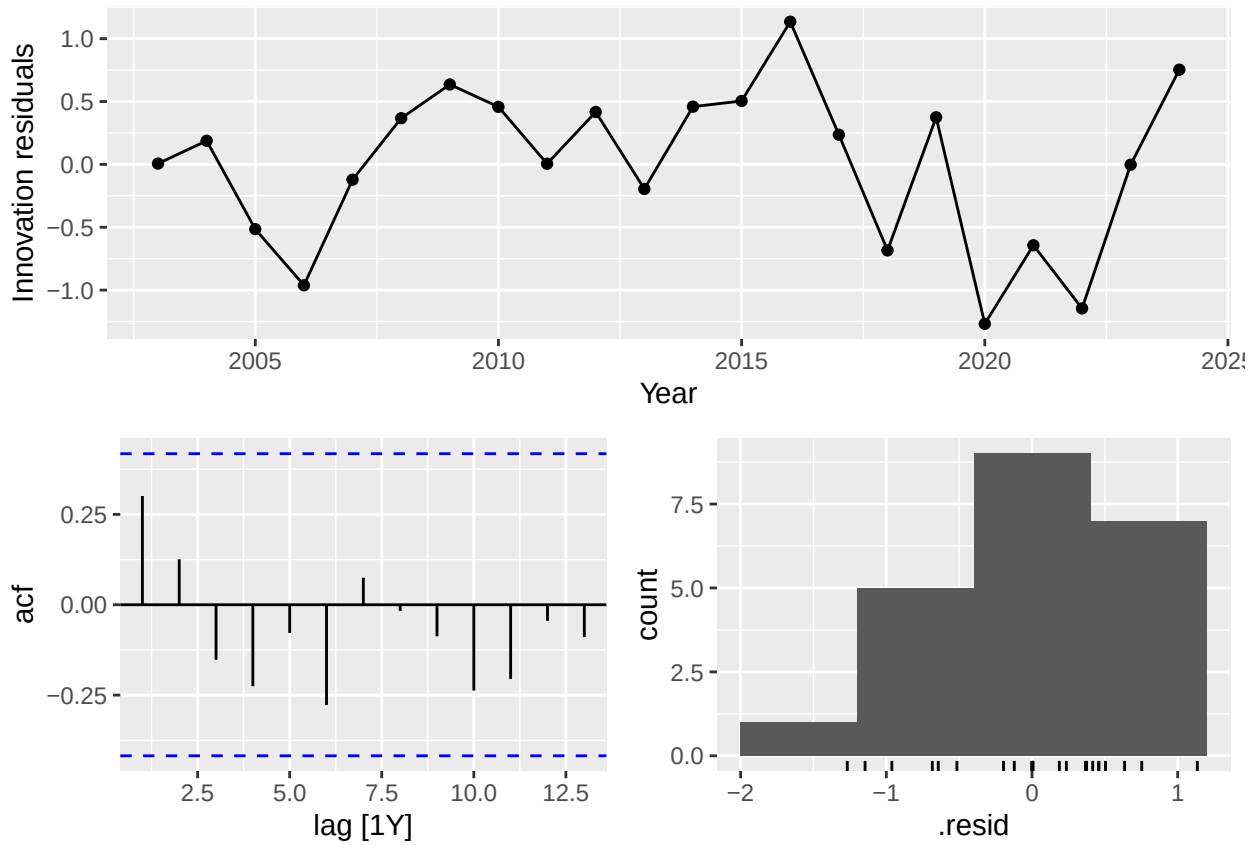



```r
library(fabletools)
library(dplyr)
library(ggplot2)
```

```r
gg_tsresiduals(fit_ft)
```




```
gg_tsresiduals(fit_pt)
```




```r
yearly_wide <- df_long %>%
  group_by(Year, Type) %>%
  summarise(mean_retention = mean(Retention, na.rm = TRUE), .groups = "drop") %>%
  mutate(Type = ifelse(Type == "FT", "FullTime", "PartTime")) %>%
  pivot_wider(
    names_from = Type,
    values_from = mean_retention
  ) %>%
  arrange(Year)

yearly_wide
```

```
## # A tibble: 22 x 3
##     Year FullTime PartTime
##    <int>    <dbl>    <dbl>
##  1  2003     64.8     44.9
##  2  2004     65.2     45.1
##  3  2005     65.1     44.4
##  4  2006     64.4     44.0
##  5  2007     64.9     44.8
##  6  2008     65.6     45.3
##  7  2009     64.9     45.5
##  8  2010     65.9     45.4
##  9  2011     65.2     44.9
## 10  2012     64.9     45.3
## # i 12 more rows
```

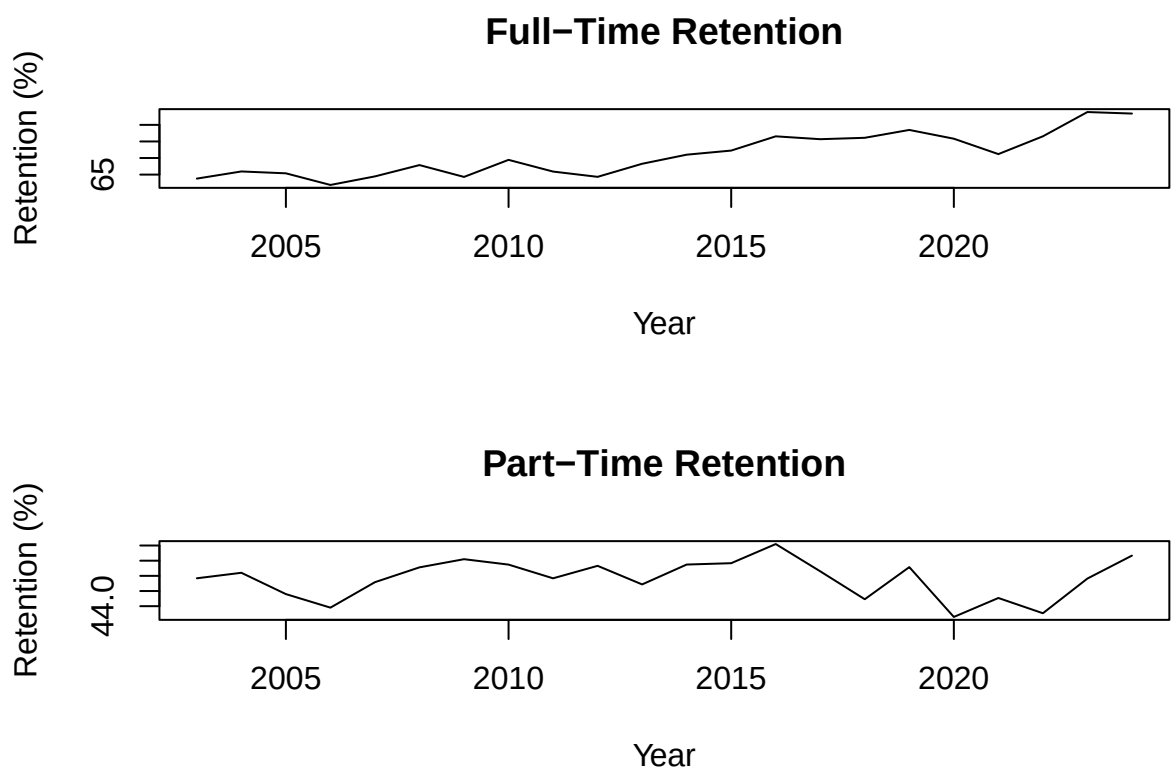
Create time-series objects

```
library(forecast)
```

```
## Warning: package 'forecast' was built under R version 4.3.3

## Registered S3 method overwritten by 'quantmod':
##   method            from
##   as.zoo.data.frame zoo
```

```
start_year <- min(yearly_wide$Year)
end_year   <- max(yearly_wide$Year)

ft_ts <- ts(yearly_wide$FullTime, start = start_year, end = end_year, frequency = 1)
pt_ts <- ts(yearly_wide$PartTime, start = start_year, end = end_year, frequency = 1)
```

```
par(mfrow = c(2,1))
plot(ft_ts, main = "Full-Time Retention", ylab = "Retention (%)", xlab = "Year")
plot(pt_ts, main = "Part-Time Retention", ylab = "Retention (%)", xlab = "Year")
```



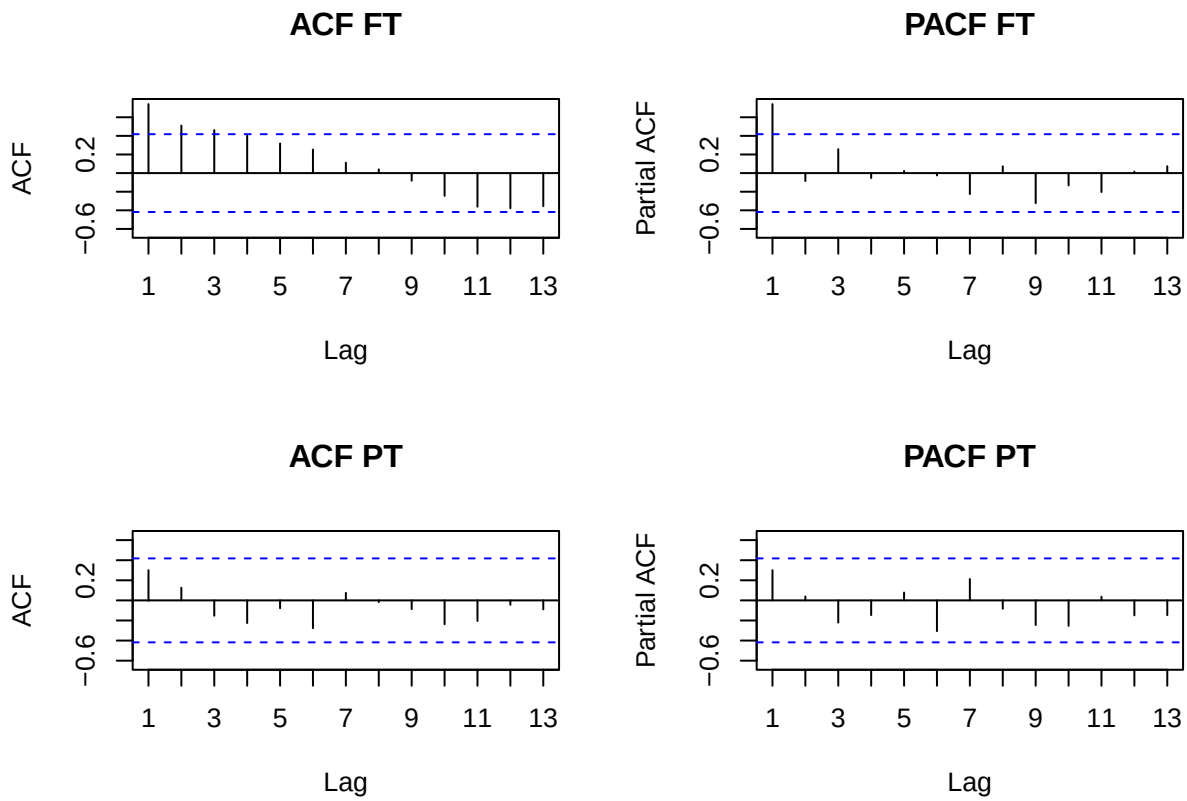

```
par(mfrow = c(1,1))
```

Plot the two series

```r
library(tseries)
```

```
## Warning: package 'tseries' was built under R version 4.3.3
```

```r
par(mfrow = c(2,2))
Acf(ft_ts, main = "ACF FT")
Pacf(ft_ts, main = "PACF FT")
Acf(pt_ts, main = "ACF PT")
Pacf(pt_ts, main = "PACF PT")
```




```r
par(mfrow = c(1,1))

adf_ft <- adf.test(ft_ts)
adf_pt <- adf.test(pt_ts)

adf_ft
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  ft_ts
## Dickey-Fuller = -2.5228, Lag order = 2, p-value = 0.3732
## alternative hypothesis: stationary
```

```r
adf_pt
```

```
##
##  Augmented Dickey-Fuller Test
##
## data:  pt_ts
## Dickey-Fuller = -2.4928, Lag order = 2, p-value = 0.3847
## alternative hypothesis: stationary
```

Stationarity + ACF/PACF

```r
fit_ft <- auto.arima(ft_ts)
fit_pt <- auto.arima(pt_ts)

summary(fit_ft)
```

```
## Series: ft_ts
## ARIMA(0,1,0)
##
## sigma^2 = 0.477:  log likelihood = -22.02
## AIC=46.04   AICc=46.25   BIC=47.08
##
## Training set error measures:
##                     ME      RMSE       MAE       MPE      MAPE      MASE
## Training set 0.1813338 0.6747384 0.5695143 0.2668362 0.8587501 0.9595047
##                    ACF1
## Training set -0.1445908
```

```r
summary(fit_pt)
```

```
## Series: pt_ts
## ARIMA(0,0,0) with non-zero mean
##
## Coefficients:
##          mean
##       44.9135
## s.e.   0.1327
##
## sigma^2 = 0.4057:  log likelihood = -20.78
## AIC=45.56   AICc=46.19   BIC=47.74
##
## Training set error measures:
##                        ME      RMSE       MAE         MPE     MAPE      MASE
## Training set 7.105427e-15 0.6222843 0.5034591 -0.01933129 1.124894 0.7855722
##                   ACF1
## Training set 0.3008119
```

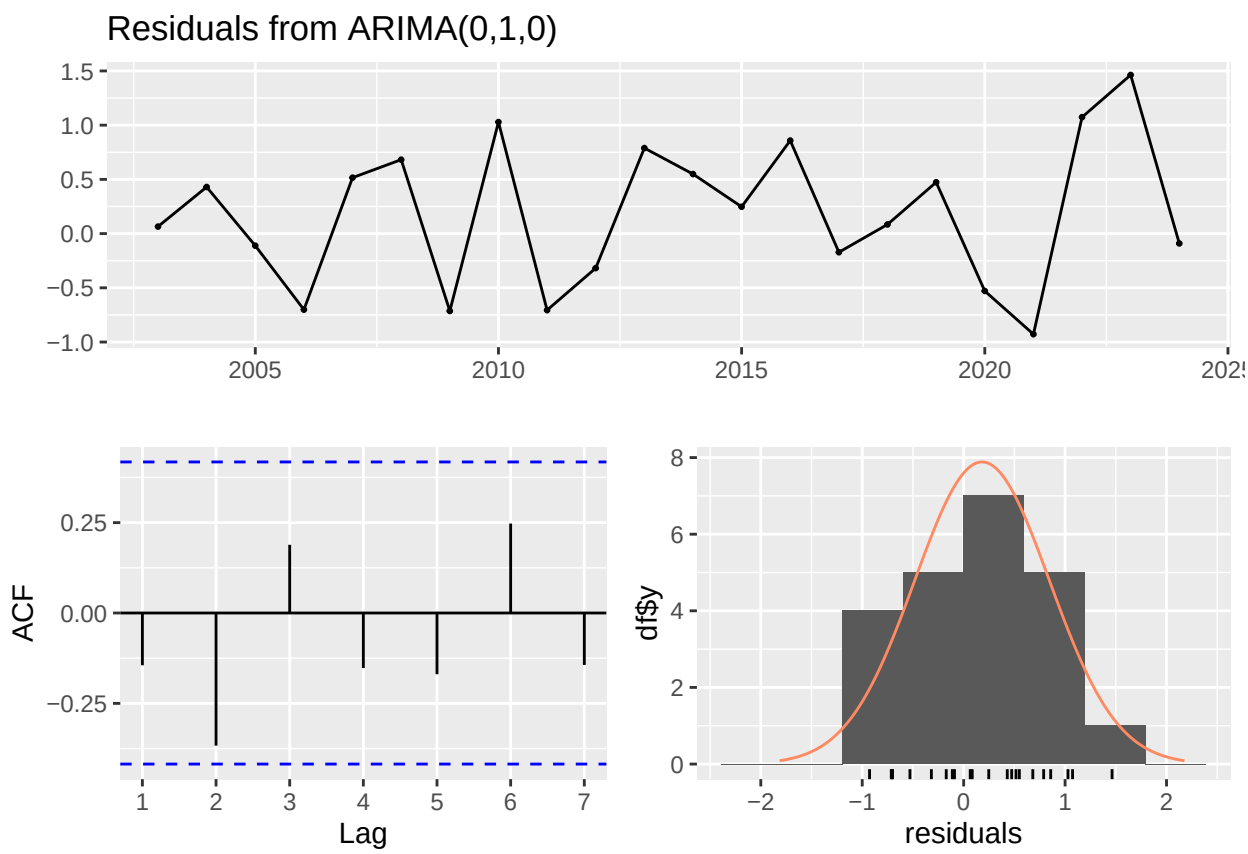
```r
AIC(fit_ft, fit_pt)
```

```
##        df      AIC
## fit_ft  1 46.03948
## fit_pt  2 45.56153
```

```r
BIC(fit_ft, fit_pt)
```

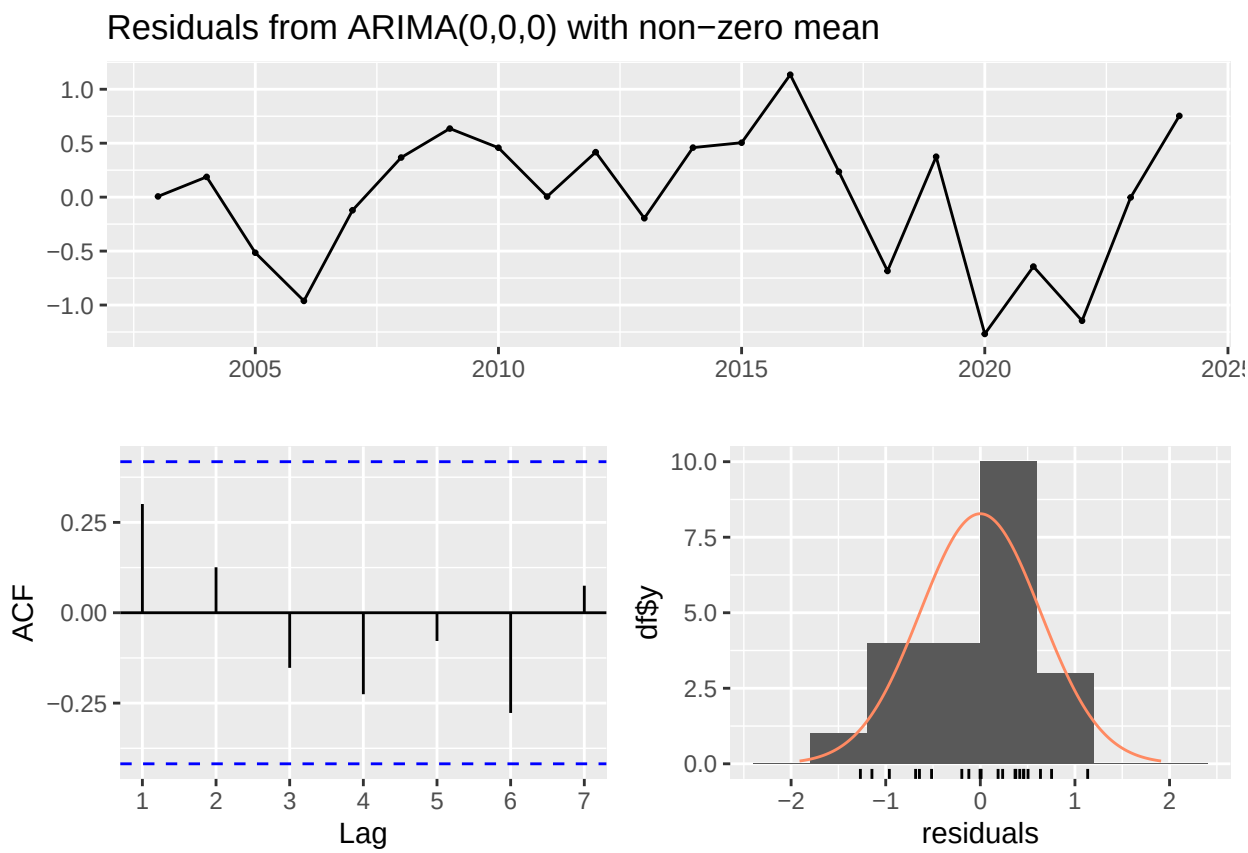
```
##        df      BIC
## fit_ft  1 47.08400
## fit_pt  2 47.74362
```

```r
checkresiduals(fit_ft)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,1,0)
## Q* = 5.7488, df = 4, p-value = 0.2187
##
## Model df: 0.   Total lags used: 4
```

```r
checkresiduals(fit_pt)
```




```
##
##  Ljung-Box test
##
## data:  Residuals from ARIMA(0,0,0) with non-zero mean
## Q* = 4.8278, df = 4, p-value = 0.3054
##
## Model df: 0.   Total lags used: 4
```

ARIMA model comparison

```r
lm_ft <- lm(FullTime ~ Year, data = yearly_wide)
lm_pt <- lm(PartTime ~ Year, data = yearly_wide)

summary(lm_ft)
```

```
##
## Call:
## lm(formula = FullTime ~ Year, data = yearly_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.30218 -0.44808  0.09679  0.41398  0.87880
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
```

```
## (Intercept) -292.16839   40.45466  -7.222 5.45e-07 ***
## Year           0.17798    0.02009   8.859 2.33e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5979 on 20 degrees of freedom
## Multiple R-squared:  0.7969, Adjusted R-squared:  0.7867
## F-statistic: 78.48 on 1 and 20 DF,  p-value: 2.33e-08
```

```r
summary(lm_pt)
```

```
##
## Call:
## lm(formula = PartTime ~ Year, data = yearly_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.21082 -0.48318  0.09255  0.42584  1.15675
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 62.603603  43.984071   1.423    0.170
## Year        -0.008786   0.021844  -0.402    0.692
##
## Residual standard error: 0.65 on 20 degrees of freedom
## Multiple R-squared:  0.008023,   Adjusted R-squared:  -0.04158
## F-statistic: 0.1618 on 1 and 20 DF,  p-value: 0.6918
```

Linear trend comparison

```r
lm_ft <- lm(FullTime ~ Year, data = yearly_wide)
lm_pt <- lm(PartTime ~ Year, data = yearly_wide)

summary(lm_ft)
```

```
##
## Call:
## lm(formula = FullTime ~ Year, data = yearly_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.30218 -0.44808  0.09679  0.41398  0.87880
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept) -292.16839   40.45466  -7.222 5.45e-07 ***
## Year           0.17798    0.02009   8.859 2.33e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.5979 on 20 degrees of freedom
## Multiple R-squared:  0.7969, Adjusted R-squared:  0.7867
## F-statistic: 78.48 on 1 and 20 DF,  p-value: 2.33e-08
```

```r
summary(lm_pt)
```

```
##
## Call:
## lm(formula = PartTime ~ Year, data = yearly_wide)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.21082 -0.48318  0.09255  0.42584  1.15675
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) 62.603603  43.984071   1.423    0.170
## Year        -0.008786   0.021844  -0.402    0.692
##
## Residual standard error: 0.65 on 20 degrees of freedom
## Multiple R-squared:  0.008023,   Adjusted R-squared:  -0.04158
## F-statistic: 0.1618 on 1 and 20 DF,  p-value: 0.6918
```

Var model

```r
library(vars)
```

```
## Loading required package: MASS

##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select

## Loading required package: strucchange

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following object is masked from 'package:tsibble':
##
##     index

## The following objects are masked from 'package:base':
##
##     as.Date, as.Date.numeric

## Loading required package: sandwich

##
## Attaching package: 'strucchange'
```

```
## The following object is masked from 'package:stringr':
##
##     boundary

## Loading required package: urca

## Loading required package: lmtest

##
## Attaching package: 'vars'

## The following object is masked from 'package:fable':
##
##     VAR
```

```
both_ts <- cbind(
  FullTime  = ft_ts,
  PartTime  = pt_ts
)

lagselect <- VARselect(both_ts, lag.max = 5, type = "const")
lagselect
```

```
## $selection
## AIC(n)  HQ(n)  SC(n) FPE(n)
##      4      4      4      4
##
## $criteria
##                 1          2          3          4          5
## AIC(n) -1.1318798 -1.1061805 -1.5752187 -1.8356107 -1.7134356
## HQ(n)  -1.1026481 -1.0574610 -1.5070115 -1.7479157 -1.6062529
## SC(n)  -0.8378045 -0.6160550 -0.8890430 -0.9533848 -0.6351595
## FPE(n)  0.3248433  0.3428784  0.2296176  0.2027163  0.2949997
```

```
p_chosen <- lagselect$selection["AIC(n)"]
```

```
var_fit <- VAR(both_ts, p = p_chosen, type = "const")
summary(var_fit)
```

```
##
## VAR Estimation Results:
## =========================
## Endogenous variables: FullTime, PartTime
## Deterministic variables: const
## Sample size: 18
## Log Likelihood: -18.256
## Roots of the characteristic polynomial:
## 1.037 0.7855 0.7855 0.7398 0.7398 0.6007 0.6007 0.04928
## Call:
## VAR(y = both_ts, p = p_chosen, type = "const")
##
```

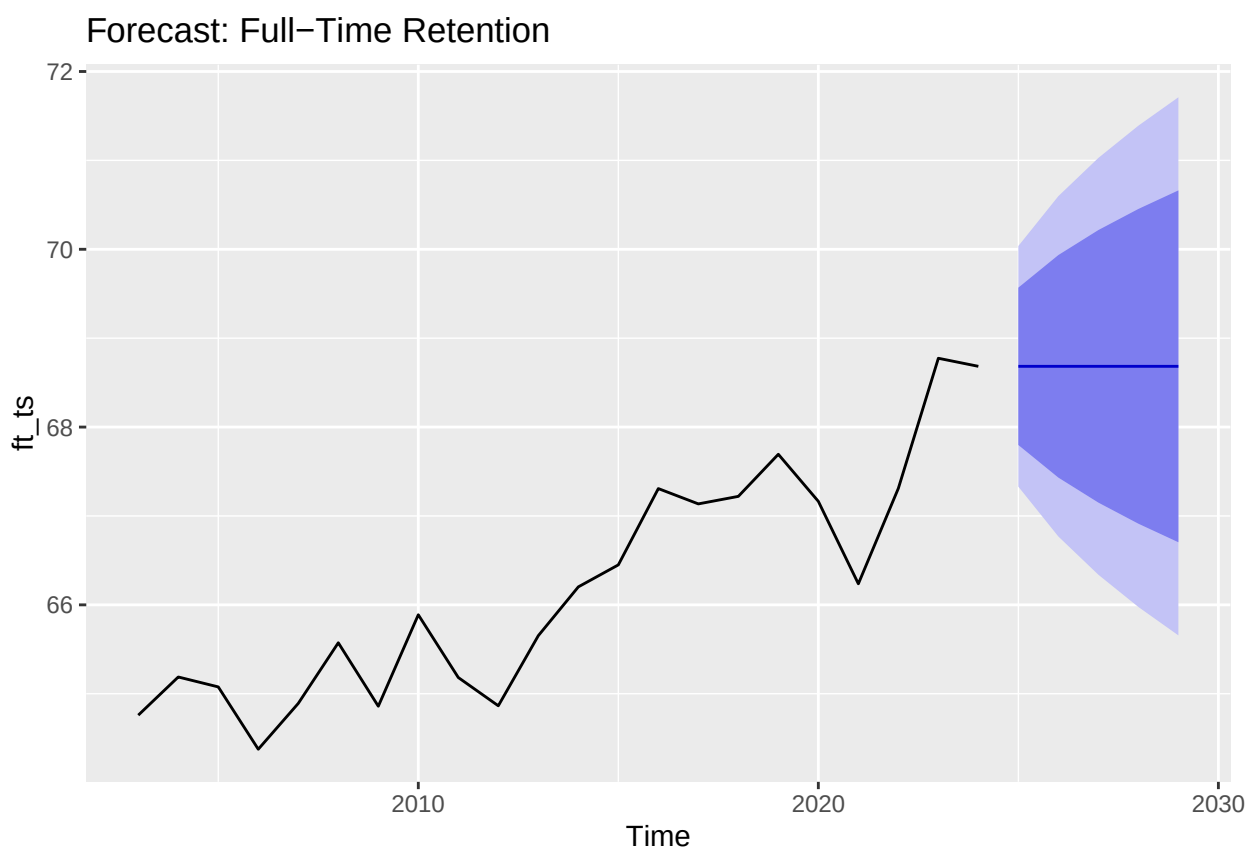
```
##
## Estimation results for equation FullTime:
## =========================================
## FullTime = FullTime.l1 + PartTime.l1 + FullTime.l2 + PartTime.l2 + FullTime.l3 + PartTime.l3 + FullT
##
##             Estimate Std. Error t value Pr(>|t|)
## FullTime.l1  0.81829    0.40318   2.030    0.073 .
## PartTime.l1 -0.02013    0.47559  -0.042    0.967
## FullTime.l2 -0.14469    0.44674  -0.324    0.753
## PartTime.l2 -0.26385    0.40290  -0.655    0.529
## FullTime.l3  0.50790    0.43498   1.168    0.273
## PartTime.l3 -0.10628    0.42691  -0.249    0.809
## FullTime.l4 -0.25298    0.56089  -0.451    0.663
## PartTime.l4 -0.05898    0.41142  -0.143    0.889
## const       25.17551   37.75733   0.667    0.522
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.8119 on 9 degrees of freedom
## Multiple R-Squared: 0.7704,  Adjusted R-squared: 0.5662
## F-statistic: 3.774 on 8 and 9 DF,  p-value: 0.03208
##
##
## Estimation results for equation PartTime:
## =========================================
## PartTime = FullTime.l1 + PartTime.l1 + FullTime.l2 + PartTime.l2 + FullTime.l3 + PartTime.l3 + FullT
##
##               Estimate Std. Error t value Pr(>|t|)
## FullTime.l1   0.626706   0.247095   2.536  0.03190 *
## PartTime.l1  -0.460443   0.291471  -1.580  0.14863
## FullTime.l2  -0.204528   0.273787  -0.747  0.47411
## PartTime.l2  -0.007754   0.246922  -0.031  0.97563
## FullTime.l3  -0.388149   0.266580  -1.456  0.17936
## PartTime.l3   0.009466   0.261637   0.036  0.97193
## FullTime.l4  -0.542952   0.343747  -1.580  0.14868
## PartTime.l4  -0.101961   0.252143  -0.404  0.69537
## const       103.498203  23.139985   4.473  0.00155 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##
## Residual standard error: 0.4976 on 9 degrees of freedom
## Multiple R-Squared: 0.6906,  Adjusted R-squared: 0.4156
## F-statistic: 2.511 on 8 and 9 DF,  p-value: 0.09605
##
##
##
## Covariance matrix of residuals:
##          FullTime PartTime
## FullTime   0.6592   0.2428
## PartTime   0.2428   0.2476
##
## Correlation matrix of residuals:
```

```
##          FullTime PartTime
## FullTime   1.0000   0.6011
## PartTime   0.6011   1.0000
```

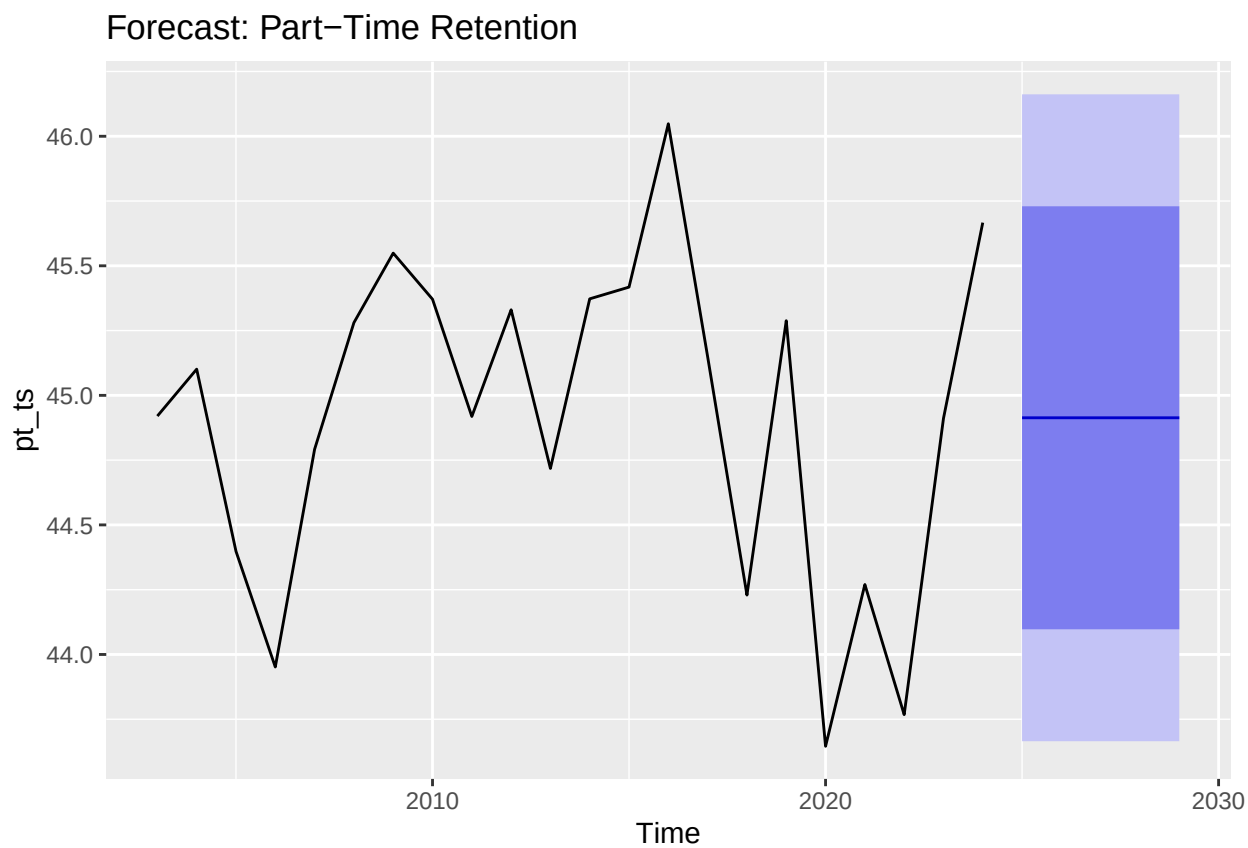
```r
serial_test <- serial.test(var_fit, lags.pt = 5, type = "PT.asymptotic")
serial_test
```

```
##
##  Portmanteau Test (asymptotic)
##
## data:  Residuals of VAR object var_fit
## Chi-squared = 11.167, df = 4, p-value = 0.02475
```

```r
# Forecast next 5 years
fc_ft <- forecast(fit_ft, h = 5)
fc_pt <- forecast(fit_pt, h = 5)

autoplot(fc_ft) + ggtitle("Forecast: Full-Time Retention")
```




```r
autoplot(fc_pt) + ggtitle("Forecast: Part-Time Retention")
```



ETS section:

```
ets_ft <- ets(ft_ts)
ets_pt <- ets(pt_ts)

summary(ets_ft)
```

```
## ETS(M,N,N)
##
## Call:
## ets(y = ft_ts)
##
##   Smoothing parameters:
##     alpha = 0.8483
##
##   Initial states:
##     l = 64.8123
##
##   sigma:  0.0106
##
##      AIC     AICc      BIC
## 56.37768 57.71101 59.65080
##
## Training set error measures:
##                     ME     RMSE       MAE       MPE      MAPE      MASE
## Training set 0.2061829 0.671797 0.5574577 0.3031706 0.8398468 0.9391921
```

```
##                      ACF1
## Training set -0.05433757
```

```r
summary(ets_pt)
```

```
## ETS(M,N,N)
##
## Call:
## ets(y = pt_ts)
##
##   Smoothing parameters:
##     alpha = 1e-04
##
##   Initial states:
##     l = 44.9135
##
##   sigma:  0.0145
##
##      AIC     AICc      BIC
## 53.13333 54.46666 56.40645
##
## Training set error measures:
##                         ME      RMSE       MAE         MPE     MAPE      MASE
## Training set -7.405647e-05 0.6223154 0.5034592 -0.01949719 1.124897 0.7855723
##                    ACF1
## Training set 0.3008087
```

```r
train_ft <- window(ft_ts, start = c(2003, 1), end = c(2017, 1))
test_ft  <- window(ft_ts, start = c(2018, 1), end = c(2024, 1))

train_pt <- window(pt_ts, start = c(2003, 1), end = c(2017, 1))
test_pt  <- window(pt_ts, start = c(2018, 1), end = c(2024, 1))


ets_ft <- ets(train_ft)
ets_pt <- ets(train_pt)

ets_ft_fc <- forecast(ets_ft, h = length(test_ft))
ets_pt_fc <- forecast(ets_pt, h = length(test_pt))
```

```r
accuracy(ets_ft_fc, test_ft)
```

```
##                      ME      RMSE       MAE       MPE      MAPE      MASE
## Training set 0.2060463 0.5757196 0.5011600 0.3074354 0.7630706 0.8969855
## Test set     0.4718241 0.9556075 0.7217249 0.6831391 1.0604231 1.2917568
##                   ACF1 Theil's U
## Training set -0.163825        NA
## Test set      0.378486  1.155431
```

```r
accuracy(ets_pt_fc, test_pt)
```

```
##                       ME      RMSE       MAE        MPE      MAPE      MASE
```

```
## Training set  0.1207119 0.4844877 0.4248893  0.2577266 0.9422721 0.8732999
## Test set     -0.8219243 1.0863762 0.9090637 -1.8711800 2.0619963 1.8684521
##                     ACF1 Theil's U
## Training set  0.15500923        NA
## Test set     -0.09065911  1.049819
```